KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 9 STANDARD

CHAPTER 10

YEAR 9 MATHEMATICS

Surface area and volume

10 lessons · two pages each · Easy, Medium, then Mixed practice

10.1 Metric units
10.2 Precision and absolute error
10.3 Percentage error
10.4 Perimeter and area of composite shapes
10.5 Areas of quadrilaterals
10.6 Circumference and arc length
10.7 Areas of circles and sectors
10.8 Surface area of a prism
10.9 Surface area of a cylinder
10.10 Volumes of prisms and cylinders

NAME _____ CLASS _____ TEACHER _____

Artwork: a silver spiral on the western edge of Seahorse Valley, coloured in purple and silver, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **two pages**. Read the **learning intention** and tick each **success criterion** when you can do it.

2. Page 1 is **Easy** and page 2 is **Medium**. Each page starts with an example, then questions for you.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. Work down the box one round at a time. Answers go on the line; show working in the boxes.

3. **Mixed practice** ends page 2: today's skill mixed with earlier lessons. The tag on each question (like **10.2** or **Year 8**) says where it comes from.

4. Stuck? Read the **Remember** line at the top of the blue box, then look back at the example you copied, or at the lesson in the tag.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
10.1	Metric units	3		☐
10.2	Precision and absolute error	5		☐
10.3	Percentage error	7		☐
10.4	Perimeter and area of composite shapes	9		☐
10.5	Areas of quadrilaterals	11		☐
10.6	Circumference and arc length	13		☐
10.7	Areas of circles and sectors	15		☐
10.8	Surface area of a prism	17		☐
10.9	Surface area of a cylinder	19		☐
10.10	Volumes of prisms and cylinders	21		☐

In this chapter you will

- convert between metric units, including very large and very small units
- find the precision, absolute error and percentage error of a measurement
- find the perimeter and area of composite shapes and quadrilaterals
- find the circumference, arc length and area of circles, sectors and composite shapes
- find the surface area of prisms and cylinders
- find the volume and capacity of prisms and cylinders



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a silver spiral on the western edge of Seahorse Valley, coloured in purple and silver. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON 10.1

Metric units

Date _____

Learning intention
We are learning to convert between metric units, and between units of time and digital storage.

Success criteria
- ☐ I can convert units of length, mass and capacity
- ☐ I can convert units of time
- ☐ I can use the prefixes mega-, giga- and micro-

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 **Convert each one. To a smaller unit, multiply. To a bigger unit, divide.**

a 8 km to m

b 250 cm to m

c 3.5 kg to g

d 4500 mL to L

YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER 1 km = 1000 m · 1 m = 100 cm · 1 cm = 10 mm · 1 t = 1000 kg · 1 kg = 1000 g · 1 kL = 1000 L · 1 L = 1000 mL

ROUND 1 Questions 1–8: convert each length.

1. 6 km to m
2. 3 000 m to km
3. 4 m to cm
4. 700 cm to m
5. 9 cm to mm
6. 120 mm to cm
7. 2.5 km to m
8. 45 cm to m

ROUND 2 Questions 9–16: convert each mass.

9. 5 kg to g
10. 2 000 g to kg
11. 3 t to kg
12. 6 500 kg to t
13. 0.8 kg to g
14. 750 g to kg
15. 4 000 mg to g
16. 1.2 t to kg

ROUND 3 Questions 17–24: convert each capacity.

17. 7 L to mL
18. 3 000 mL to L
19. 2 kL to L
20. 4 500 L to kL
21. 1.5 L to mL
22. 250 mL to L
23. 0.6 kL to L
24. 80 mL to L

ROUND 4 Questions 25–32: convert each one. Some take two steps.

25. 4 km to cm
26. 3 m to mm
27. 60 000 cm to km
28. 5 000 mm to m
29. 2 kg to mg
30. 1 kL to mL
31. 0.5 km to cm
32. 80 mm to m

Level 2 · Building up

Medium · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Convert each one. Time is not metric: 1 h = 60 min and 1 min = 60 s.**

a 3.5 h to min

b 150 s to min

c 4.2 GB to MB

d 2 500 000 µm to m

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

18 QUESTIONS

REMEMBER mega- (M) = 1 000 000 · giga- (G) = 1 000 000 000 · tera- (T) = 1000 giga- · micro- (µ) = one-millionth · 1 s = 1000 ms

ROUND 1 Questions 1–8: convert each time.

1. 5 min to s
2. 4 h to min
3. 180 s to min
4. 2.5 h to min
5. 90 min to h
6. 3 days to h
7. 4 000 ms to s
8. 1.5 min to s

ROUND 2 Questions 9–16: convert using the prefixes.

9. 3 GB to MB
10. 4 500 MB to GB
11. 2.4 TB to GB
12. 34 000 B to kB
13. 2.6 Mm to m
14. 0.7 s to ms
15. 3 000 000 µm to m
16. 5 ms to µs

ROUND 3 Questions 17–18: solve each problem. Show your working.

17. A phone has 64 GB of free space. Each video is 800 MB. How many videos fit?

18. A movie runs for 135 minutes. Write this in hours and minutes.

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

8 QUESTIONS

1. Convert 0.75 km to m. [10.1]
2. Find the area of a rectangle 8 cm long and 5 cm wide. [Year 8]
3. Which is longer: 1.2 m or 115 cm? [10.1]
4. Round 3.476 to one decimal place. [Year 8]
5. How many seconds are in 2 hours? [10.1]
6. Find the perimeter of a square with sides of 7 m. [Year 8]
7. A 2 L jug is filled with 400 mL cups of water. How many cups does it take? [10.1]
8. Convert 2.5 GB to MB. [10.1]

LESSON 10.2

Precision and absolute error

Date _____

Learning intention
We are learning to find the precision and absolute error of a measuring instrument or a measurement.

Success criteria
- ☐ I can find the precision: the smallest unit on the scale
- ☐ I can find the absolute error: $\frac{1}{2}$ of the precision
- ☐ I can find the smallest and largest possible true value

Level 1 · Getting started

Easy · 20 questions

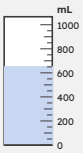
WE DO With your teacher: copy what they write on the board.

Example 1 Find the precision and the absolute error of each.

a This ruler



b This jug



c A path 24 m long, to the nearest metre

d A bag of flour with a mass of 3.6 kg

YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

20 QUESTIONS

REMEMBER The precision of a measurement is the place value of its last digit: 3.6 kg is to the nearest 0.1 kg. Absolute error = precision ÷ 2.

ROUND 1 Questions 1–8: write the precision of each measurement.

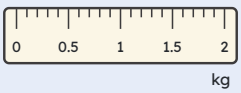
1. 7 m
2. 3.6 cm
3. 12.45 kg
4. 58 mm
5. 2.8 L
6. 0.75 t
7. 104 g
8. 9.3 s

ROUND 2 Questions 9–16: write the absolute error of each measurement.

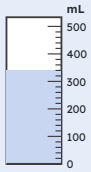
9. 15 m
10. 6.2 kg
11. 43 mL
12. 8.25 m
13. 125 cm
14. 0.4 L
15. 19.75 s
16. 312 g

ROUND 3 Questions 17–20: write the precision and the absolute error of each scale.

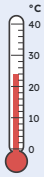
17. Kitchen scales



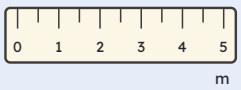
18. Jug



19. Thermometer



20. Tape measure



Level 2 · Building up

Medium · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Limits: from measurement − absolute error to measurement + absolute error.**

a A plank is 2.45 m long. Find its precision and absolute error.

b Write the limits of the plank's length in part a.

c Jo is 162 cm tall, to the nearest cm. Write the limits.

d A jug marked every 50 mL shows 650 mL. Write the limits.

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

18 QUESTIONS

REMEMBER Smallest = measurement − absolute error · Largest = measurement + absolute error

ROUND 1 — Questions 1–8: write the absolute error of each measurement.

1. 3.25 kg
2. 0.8 m
3. 14.6 L
4. 7.125 km
5. 0.04 g
6. 2.05 m
7. 120.5 cm
8. 6.375 t

ROUND 2 — Questions 9–16: write the smallest and largest possible true value.

9. 24 m
10. 6.4 cm
11. 153 g
12. 3.25 kg
13. 9.8 s
14. 72 kg
15. 1.6 L
16. 0.45 m

ROUND 3 — Questions 17–18: solve each problem. Show your working.

17. Kai weighs 250 g of flour on scales marked every 10 g. Write the limits of the mass.

18. A rope is 12.5 m long, to the nearest 0.1 m. Could it really be 12.56 m? Explain.

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

8 QUESTIONS

1. What is the precision of 4.75 m? [10.2]
2. Convert 3.2 km to m. [10.1]
3. What is the absolute error of a ruler marked in mm? [10.2]
4. Find 5% of 60. [Year 8]
5. Write the limits of 18 cm. [10.2]
6. Convert 750 mL to L. [10.1]
7. Scales are marked every 20 g. Absolute error? [10.2]
8. What is the absolute error of 6.3 L? [10.2]

LESSON 10.3

Percentage error

Date _____

Learning intention
We are learning to calculate the percentage error of a measurement, and use it to compare accuracy.

Success criteria
- ☐ I can find the absolute error of a measurement
- ☐ I can calculate percentage error = absolute error ÷ measurement × 100%
- ☐ I can decide which measurement is more accurate

Level 1 · Getting started

Easy · 24 questions

WE DO With your teacher: copy what they write on the board.

Example 1 **Find the percentage error of each measurement.**

a 20 m, to the nearest metre

b 4 kg, to the nearest kilogram

c 2.5 L

d 50 cm, to the nearest centimetre

YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER Step 1: absolute error = precision ÷ 2. Step 2: percentage error = absolute error ÷ measurement × 100%.

ROUND 1 Questions 1–8: find the percentage error. Each is to the nearest whole unit.

1. 10 m
2. 25 cm
3. 50 kg
4. 4 m
5. 20 s
6. 100 g
7. 5 L
8. 40 mm

ROUND 2 Questions 9–16: find the percentage error of each measurement.

9. 2.5 m
10. 0.5 kg
11. 1.6 L
12. 12.5 cm
13. 0.2 km
14. 0.4 t
15. 6.25 s
16. 0.8 m

ROUND 3 Questions 17–24: find the percentage error, correct to one decimal place.

17. 7 m
18. 15 cm
19. 9 kg
20. 30 s
21. 3.7 L
22. 0.6 m
23. 45 g
24. 2.3 km

Level 2 · Building up

Medium · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Use a calculator. Give each answer correct to 2 decimal places.**

a A yacht is 24 m long. Find its absolute error and percentage error.

b A parcel has a mass of 3.25 kg. Find its percentage error.

c A jug marked every 25 mL shows 375 mL. Find the percentage error.

d Which of a, b and c is the most accurate? Why?

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

14 QUESTIONS

REMEMBER The smaller the percentage error, the more accurate the measurement. Use the instrument's precision when it is given.

ROUND 1 Questions 1–8: find the percentage error, correct to 2 decimal places.

1. 24 m
2. 3.6 cm
3. 12.45 kg
4. 175 cm
5. 0.8 L
6. 2.35 m
7. 48 s
8. 9.5 kg

ROUND 2 Questions 9–12: find the percentage error, correct to 2 decimal places.

9. 350 g on scales marked every 10 g
10. 75 mL in a jug marked every 25 mL
11. 38°C on a thermometer marked every 2°
12. 1.2 m on a tape marked every 5 cm

ROUND 3 Questions 13–14: solve each problem. Show your working.

13. Mia's pencil is 15 cm long and she is 168 cm tall, both to the nearest cm. Which is more accurate?
14. A 100 m sprint is timed as 12.4 s. Find the percentage error of the time.

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

8 QUESTIONS

1. Find the percentage error of 8 m, to the nearest metre. [10.3]
2. Write the limits of 5.3 kg. [10.2]
3. A scale marked every 20 g shows 400 g. Find the percentage error. [10.3]
4. Convert 4500 g to kg. [10.1]
5. Find the percentage error of 0.5 L. [10.3]
6. Which is more accurate: 10 m or 100 m, each to the nearest metre? [10.3]
7. Find the area of a triangle with base 10 cm and height 6 cm. [Year 8]
8. Find the percentage error of 2.5 cm. [10.3]

LESSON 10.4

Perimeter and area of composite shapes

Date _____

Learning intention
We are learning to find the perimeter and area of shapes made from rectangles and triangles.

Success criteria
- ☐ I can find missing side lengths, then the perimeter
- ☐ I can find an area by splitting a shape into rectangles and triangles
- ☐ I can find a shaded area by subtracting

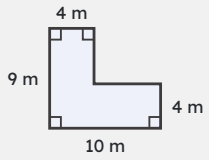
Level 1 · Getting started

Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

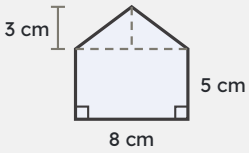
Example 1 Find the perimeter or area. Find any missing sides first.

a Perimeter

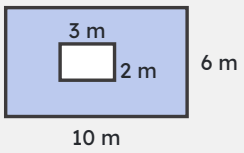


b Area of the shape in part a

c Area



d Shaded area

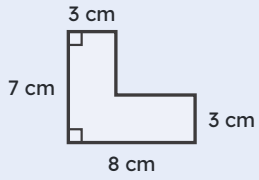


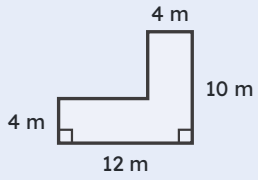
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

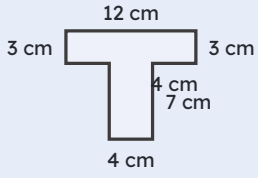
12 QUESTIONS

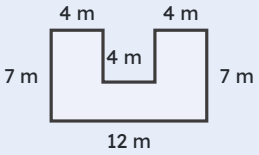
REMEMBER Rectangle A = l × w · Triangle A = ½ × b × h · Shaded area = whole shape − the white part · Perimeter: add every side.

ROUND 1 Questions 1–4: find the perimeter of each shape.

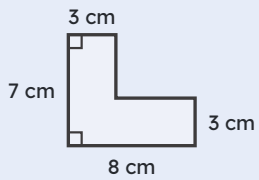
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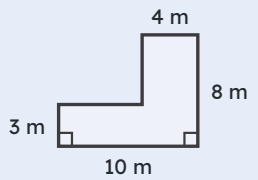
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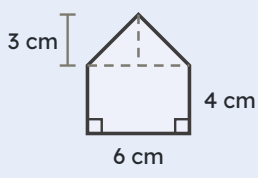
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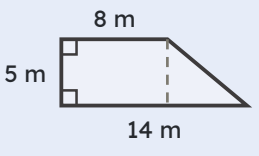
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ROUND 2 Questions 5–8: find the area of each shape.

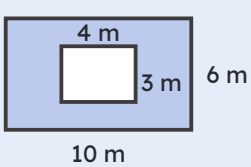
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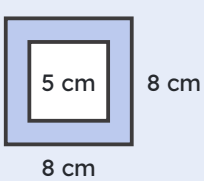
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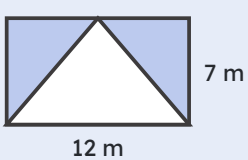
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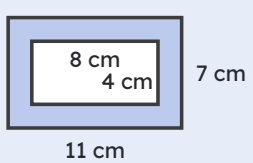
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ROUND 3 Questions 9–12: find each shaded area.

9.


10.


11.


12.


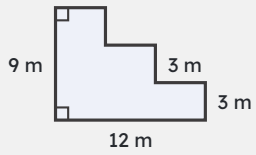
Level 2 · Building up

Medium · 8 questions

WE DO With your teacher: copy what they write on the board.

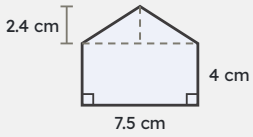
Example 2 **Find the perimeter or area. Split the shape, or subtract.**

a Perimeter

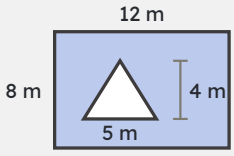


b Area of the shape in part a. Each step is 4 m wide.

c Area



d Shaded area



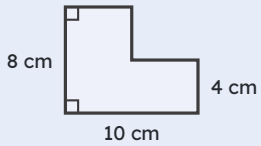
YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line.

8 QUESTIONS

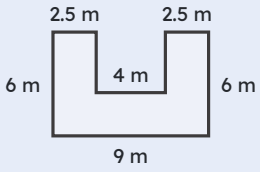
REMEMBER Missing sides: use the opposite sides (across the shape the lengths add up). Split into rectangles and triangles, or take away.

ROUND 1 Questions 1–4: find the perimeter of each shape.

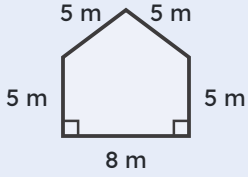
1



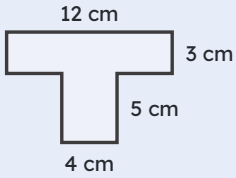
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3

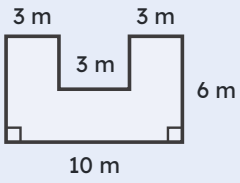


4 It is symmetrical.

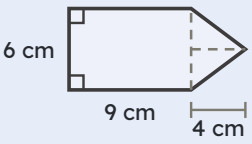


ROUND 2 Questions 5–8: find the area of each shape.

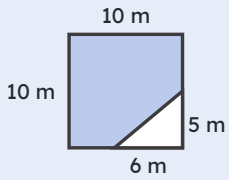
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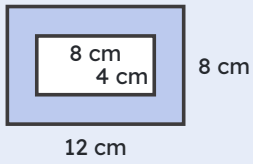
6



7



8



MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

8 QUESTIONS

1. A rectangle is 12 m by 5 m. Find its perimeter. [10.4]
2. Find the percentage error of 40 cm, to the nearest cm. [10.3]
3. A 12 m by 8 m lawn has a 3 m by 2 m garden bed. Area of lawn? [10.4]
4. Convert 2.4 m to cm. [10.1]
5. A triangle has base 9 m and height 4 m. Find its area. [10.4]
6. Write the limits of 7.2 m. [10.2]
7. An L-shape is a 6 m by 2 m and a 3 m by 2 m rectangle. Area? [10.4]
8. A square has an area of 49 cm². Find its perimeter. [10.4]

LESSON 10.5

Areas of quadrilaterals

Date _____

Learning intention
We are learning to find the area of a parallelogram, a trapezium, a rhombus and a kite.

Success criteria
- ☐ I can use A = bh for a parallelogram
- ☐ I can use A = $\frac{1}{2}$(a + b)h for a trapezium
- ☐ I can use A = $\frac{1}{2}$xy for a rhombus or a kite

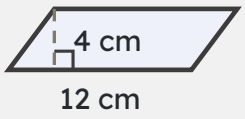
Level 1 · Getting started

Easy · 12 questions

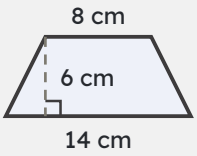
WE DO With your teacher: copy what they write on the board.

Example 1 **Find the area of each shape. Write the formula first.**

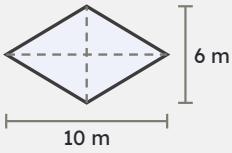
a Parallelogram



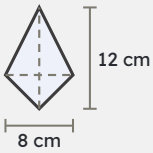
b Trapezium



c Rhombus



d Kite



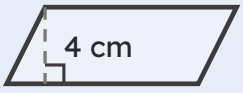
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

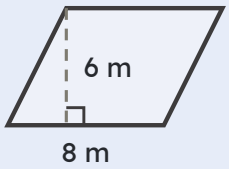
REMEMBER h is the perpendicular height (the dashed line) · a and b are the parallel sides · x and y are the diagonals.

ROUND 1 Questions 1–4: find the area of each parallelogram.

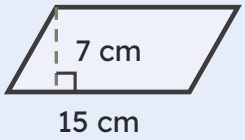
1.



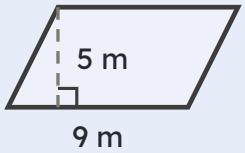
2.



3.

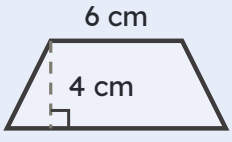


4.

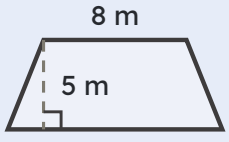


ROUND 2 Questions 5–8: find the area of each trapezium.

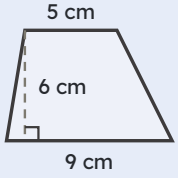
5.



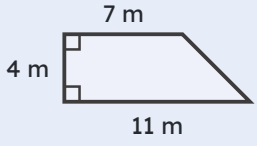
6.



7.

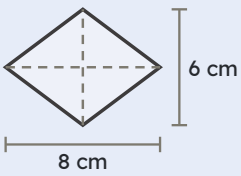


8.

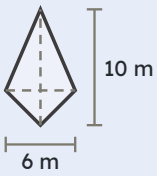


ROUND 3 Questions 9–12: find the area of each rhombus or kite.

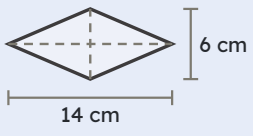
9.



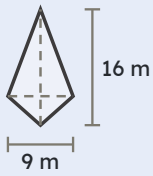
10.



11.



12.



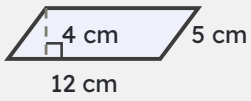
Level 2 · Building up

Medium · 8 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Choose the right formula. Some measurements are not needed.**

a Area



b A trapezium has parallel sides 7.5 m and 4.5 m, 3.2 m apart. Find its area.

c A parallelogram has an area of 54 cm² and a base of 9 cm. Find its height.

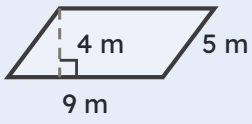
d A kite has diagonals of 1.2 m and 0.8 m. Find its area.

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 8 QUESTIONS

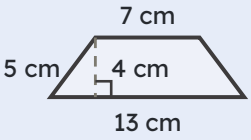
REMEMBER The slanted side is not the height · Missing length: put the numbers into the formula, then work backwards.

ROUND 1 **Questions 1–4: find the area of each shape. One length in each is not needed.**

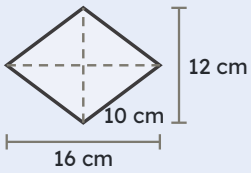
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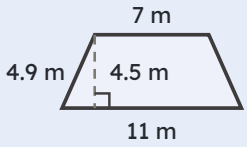
2



3



4



ROUND 2 **Questions 5–8: find the missing measurement. Show your working.**

5 A parallelogram has an area of 72 cm² and a base of 9 cm. Find its height.

6 A rhombus has an area of 24 m² and one diagonal of 8 m. Find the other diagonal.

7 A trapezium has parallel sides of 6 cm and 10 cm and an area of 40 cm². Find its height.

8 A square has diagonals of 10 cm. Use the rhombus formula to find its area.

MIXED PRACTICE **Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.** 8 QUESTIONS

1 Find the area of a parallelogram with base 7 m and height 3 m. [10.5]

2 Find the perimeter of a rectangle 9 cm by 4 cm. [10.4]

3 A trapezium has parallel sides 4 cm and 6 cm, 5 cm apart. Area? [10.5]

4 Find the percentage error of 2.5 kg. [10.3]

5 A rhombus has diagonals of 9 m and 4 m. Find its area. [10.5]

6 Convert 0.6 km to m. [10.1]

7 A kite has diagonals of 10 cm and 7 cm. Find its area. [10.5]

8 A parallelogram has an area of 30 m² and a base of 6 m. Height? [10.5]